

HARDIK JAIN

AI/ML ENGINEER · RAG & NLP SPECIALIST · B.TECH CSE (AI/ML), VIT BHOPAL — CGPA: 8.9 / 10

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SUMMARY

Pre-final year B.Tech CSE (AI/ML) student (8.9 CGPA) with hands-on research in RAG, NLP (BERT, NER), and Computer Vision, 2 published papers, 2 ML internships with quantified outcomes, and 3 production-grade AI applications shipped. Ranked global top 1,000 on [TensorTonic](#) through daily ML problem-solving AND ranked GLOBAL 5774IN TCS CODEVITA SEASON-13. Seeking to accelerate applied ML research through the Amazon ML Summer School.

TECHNICAL SKILLS

- ML & DL Frameworks: PyTorch, Scikit-learn, TensorFlow, Keras
- NLP & Generative AI: RAG (Retrieval-Augmented Generation), BERT, NER, Sentiment Analysis, LLMs, Prompt Engineering, Agentic Frameworks
- Computer Vision: YOLOv8, MobileNetV2 (Transfer Learning), OpenCV, VLMs (Moondream2, Gemini API)
- Data & Libraries: NumPy, Pandas, Matplotlib, Seaborn
- Programming: Python (Primary), Java, SQL
- Backend & Infra: FastAPI, MongoDB, MySQL, Git, Vercel, Gmail API

VIRTUAL INTERNSHIPS

- Virtual AI/ML Intern · 1M1B × IBM SkillsBuild Dec 2025 – Jan 2026
 - Built an end-to-end RAG-powered waste classification system (4 classes) grounding LLM outputs in policy documents for localised, hallucination-reduced guidance — applying Responsible AI principles throughout.
- Virtual ML Intern · Shell India Markets Pvt. Ltd JUN 2025 – JUL 2025
 - Built a 6-class waste image classifier (93% accuracy, 0.93 weighted F1) via two-phase MobileNetV2 transfer learning : frozen base → top-20-layer fine-tuning (Adam 1e-5, dropout 0.5) with augmentation on 2,527 images; ~220ms batch inference.

RESEARCH PUBLICATIONS @ IJIRT

- ResumeBERT-HIRENET : Intelligent Agentic Framework for Autonomous Recruitment Optimisation OCTOBER-2025
 - Agentic NLP platform using BERT + NER to parse unstructured resumes into structured profiles with a fairness-aware scoring algorithm (demographic parity < 0.05), eliminating bias in automated hiring.
- RailVLM — YOLOv8 + Vision Language Model Framework for Railway Track Fault Detection APRIL-2026
 - Combined YOLOv8 + VLMs (Moondream2, Gemini) for real-time, explainable railway track fault detection with human-readable diagnostics — 70% reduction in inspection report generation time.

PROJECTS

- AutoEmail-RAG : RAG-powered web application that generates context-aware Gmail reply drafts by retrieving relevant context from a user's past email history and uploaded company policy documents.
- SURAKSHA-MAPS : Building an Agentic system that monitors regulatory changes, translates them into "Measurable Action Points" (MAPs), assigns them to the correct bank departments, and autonomously validates completion.

CERTIFICATIONS

- IBM Advanced Generative AI Professional Certificate ONGOING
- Programming with Generative AI : NPTEL Aug – Oct 2025
- Applied Machine Learning in Python : University of Michigan, Coursera Nov – Dec 2025
- Python for Data Science, AI & Development : IBM / Coursera Sep – Oct 2024